MIL-B-45390A(AT)
16 March 1987
SUPERSEDING
MIL-B-45390(Ord)
29 November 1961

MILITARY SPECIFICATION

BULLDOZER KIT, EARTH-MOVING: TANK MOUNTING, M9

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulically operated, double-acting, straight-blade M9 bulldozer kits, designed for mounting on the front of fulltracked, combat tanks in the 105 MM, M60 series.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2590

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SPECIFICATIONS
MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government drawing. The following other Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

8750001	- Bulldozer, Earth-Moving, M9.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish a complete bulldozer kit, which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent bulldozer kits delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.6.1).

3.2.1 Qualified products. The contractor shall be responsible for using parts and assemblies that are listed as qualified products on the Qualified Products List (QPL) and those approved for inclusion on the QPL. The contractor's manufacturing inspection records shall specifically list all QPL items by part or drawing number(s), name of supplier(s), and number and date of the QPL. When assemblies and parts are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document with the name of supplier(s) (see 4.6.1).

3.3 Design and construction. The bulldozer kits shall be constructed and assembled into a complete assembly in accordance with drawings, specifications, and standards listed or referred to on Drawing 8750001. All parts requiring identification shall be marked in accordance with standard MIL-STD-130 (see 4.6.1 and 4.6.2).

3.3.1 Service products. Bulldozer performance shall be demonstrated using standard service products in the hydraulic system, of the following type and grade for the applicable ambient air temperature; at ambient air temperature from -10 degrees Fahrenheit (^{o}F) to +125°F, oil shall conform to grade 10 of MIL-L-2104; at ambient air temperature from -10 to -40°F, oil shall conform to MIL-L-46167 (see 4.6.1).

3.3.2 Hydraulic system. Provisions shall be made prior to initial filling to assure that oil confining areas of the hydraulic system are free of rust, scale, welding slag and spatter, or other contamination that may cause equipment failure (see 4.6.1).

3.3.3 Installation instructions. One copy of an approved set of installation instructions shall be provided with each bulldozer kit (see 4.6.2).

3.3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other in terms of installation and performance (see 4.6.1).

3.4 Performance. After the bulldozer kit is completely assembled and installed on the tank, the bulldozer, serviced as specified in 3.3.1, shall meet the performance requirements specified herein (see 4.6.3).

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3.4.1 Hydraulic operating system. The hydraulic system shall be powered by a hydraulic pump driven through a right angle drive speed reducer attachable to the transmission power take-off shaft. A four-position control valve shall direct the oil flow to the double-acting hydraulic cylinders for control of the blade. The pump when operating at 2800 revolutions per minute (rpm) shall deliver a minimum flow of 60 gallons per minute at a pressure of 1000 pounds per square inch (psi) (see 4.6.3.1).

3.4.2 Controls and linkage. Prior to initial bulldozing operation, all controls and linkage shall be operated through all ranges of operation (see 4.6.3.2).

3.4.3 Safety relief valve. Hydraulic system shall be protected from high shock pressures by a relief valve set to operate at 5000 psi (see 4.6.3.3).

3.4.4 Digging depth. Full downward pressure applied to piston shall hold the cutting edge of the blade down to not less than 9 inches below ground level. The hydraulic system shall hold the moldboard at any height within operating range, or allow moldboard to float at ground level (see 4.6.3.4).

3.4.5 Rate of lift. The hydraulic system shall lift and lower the moldboard at rates of speed listed in table I. The specified rates of speed shall be accomplished when using oil specified in 3.3.1 with an applied pressure of 1000 psi (see 4.6.3.5).

TABLE I. Rate of lift.

Engine speed (rpm) minimum	Minimum rate of lift or lower (inches per second)	
	Up	Down
1000	2.5	6.5
1500	4	6.5
2000	5	6.5
2400	5.5	6.5
2800	6	6.5

3.4.6 Environmental.

3.4.6.1 High temperature. The bulldozer shall meet the requirements of 3.4.1 through 3.4.5 during and after exposure to temperatures up to 115°F (see 4.6.1.6.1).

3.4.6.2 Low temperature. The bulldozer shall meet the requirements of 3.4.1 through 3.4.5 during and after exposure to temperatures down to -40°F (see 4.6.3.6.2).

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3.5 Identification marking. When specified (see 6.2), the bulldozer kit shall be marked as specified by the procuring activity in accordance with MIL-STD-642 (see 4.6.2).

3.5.1 Nameplate. A nameplate conforming to the applicable drawing shall be provided (see 4.6.2).

3.6 Finish. Unless otherwise specified on the applicable drawing, the Bulldozer Kit's component assemblies and parts shall be prepared and painted in accordance with MIL-STD-193 (see 4.6.2).

3.7 Workmanship. Workmanship shall be of a quality to assure that the bulldozer kit and its components are free of defects that compromise, limit, or reduce capability in the performance of its intended use. Electric leads and hydraulic lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. There shall be no loose, damaged, or missing assemblies, components, or attaching hardware. All moving parts shall operate without sticking or binding (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operation to determine that practices, methods, and procedures of the written quality assurance system plan are being properly applied, and (b) Government product inspection to measure the quality of the product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an adverse effect upon the quality of the product will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been demonstrated.

4.1.4 Qualified products. When a part or component is specified to conform to a specification having a Qualified Products List (QPL), the contractor shall make available to the Government documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity.

4.1.5 Quality assurance provisions. In the conduct of inspection, the contractor shall adhere to Quality Assurance Provisions (QAP) and General Quality Assurance Provisions (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F
b. Barometric pressure 28.5 + 2 in Mercury (Hg)
   - 3
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. First article inspection shall be performed at the location specified by the procuring activity (see 6.2). Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply bulldozer kit's that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

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4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of one bulldozer kit. Preproduction inspection shall consist of inspection as specified in table II.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select one bulldozer kit, from the first ten units produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table II.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Materials and construction	3.2 thru 3.3.2 and 3.3.4	4.6.1	X			
Defect (see 4.5.2 and table III)	3.3, 3.3.3 and 3.5 thru 3.7	4.6.2	X	X		
Performance	3.4	4.6.3	X		X	
Hydraulic operating system	3.4.1	4.6.3.1	X		X	
Controls and linkage	3.4.2	4.6.3.2	X		X	
Safety relief valve	3.4.3	4.6.3.3	X		X	
Digging depth	3.4.4	4.6.3.4	X		X	
Rate of lift	3.4.5	4.6.3.5	X			
High temperature	3.4.6.1	4.6.3.6.1	X			
Low temperature	3.4.6.2	4.6.3.6.2	X			

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4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the bulldozer kits of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 3 consecutively produced bulldozer kits that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQLs).

Classification	AQL
Major	25/100
Minor	150/100

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 25 defects/100 units	
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, not within tolerance (see 3.3).	SIE 1/
103	Right angle drive, pump, and coupling; improper assembly.	Visual
104	Front mounting brackets and moldboard; improper weld or improper assembly.	Visual
105	Headlamps and guards; improper assembly, inoperative or damaged.	Visual and Functional
106	Control assembly and guards; improper assembly, damaged, malfunction, or improper weld.	Visual and Functional
107	Installation fixture and templates; damaged or missing.	Visual and Functional
108	Hydraulic cylinders, reservoir, filter, hoses, pipes, and guards; improper assembly, malfunction, leaks or improper weld.	Visual and Functional

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TABLE III. Classification of defects – Continued.

Category	Defect	Method of examination
109	Faulty workmanship affecting performance (see 3.7).	Visual and Functional
110	Clutch, switch, lamp, circuit breaker, and leads; improper assembly, damaged, or inoperative.	Visual and Functional
Minor	AQL 150 defects/100 units	
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3).	SIE 1/
202	Identification marking, improper (see 3.3 and 3.5).	Visual
203	Technical literature; missing or incomplete (see 3.3.3).	Visual
204	Right angle drive, pump and coupling; improper installation or leaks.	Visual and Functional
205	Front mounting brackets and moldboard; improper adjustment.	Visual
206	Control assembly and guards; improper installation.	Visual and Functional
207	Hydraulic cylinders, reservoir, filter, hoses, pipes, and guards; improper installation, incorrect fluid level or type.	Visual and Functional
208	Finish; improper coverage, application, or materials (see 3.6).	Visual
209	Faulty workmanship affecting appearance (see 3.7 and 4.5.2.2.1).	Visual
210	Clutch, switch, lamp, circuit breaker, and leads; improper installation.	Visual and Functional

1/ SIE = Standard Inspection Equipment.

4.5.2.2.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a 30-day period will be added to the minor defects classification with no increase in AQLs.

4.5.3 Test. One bulldozer kit from each 50 bulldozer kits shall be subjected to the tests specified in table II.

4.5.4 Failure. Failure of any bulldozer kit to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 through 3.3.2 and 3.3.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3, 3.3.3 and 3.5 through 3.7 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Performance. To determine conformance to 3.4, the bulldozer kit shall be installed on a tank or on a simulated test unit, serviced as specified in 3.3.1 and subjected to the tests specified herein.

4.6.3.1 Hydraulic operating system. To determine conformance to 3.4.1, the bulldozer shall be operated throughout all ranges of operation, and checked for system operating pressure, hydraulic pump flow rate, and engine operating speeds.

4.6.3.2 Controls and linkage. To determine conformance to 3.4.2, the bulldozer shall be operated and the controls and linkage shall be checked for functional requirements.

4.6.3.3 Safety relief valve. To determine conformance to 3.4.3, the bulldozer shall be operated in such a manner to cause the relief valve to function.

4.6.3.4 Digging depth. To determine conformance to 3.4.4, the bulldozer shall be operated as specified and checked for digging depth and floating at ground level.

4.6.3.5 Rate of lift. To determine conformance to 3.4.5, the bulldozer shall be operated and the rate of lift for the moldboard shall be checked for all engine speeds shown in table I.

4.6.3.6 Environmental.

4.6.3.6.1 High temperature. To determine conformance to 3.4.6.1, the bulldozer shall pass the tests specified in 4.6.3.1 through 4.6.3.5 during exposure to a temperature of 115 $\pm$ 5°F. Subsequently, the bulldozer shall be returned to the room ambient conditions specified in 4.3 and shall pass the tests specified in 4.6.3.1 through 4.6.3.5.

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4.6.3.6.2 Low temperature. To determine conformance to 3.4.6.2, the bulldozer shall pass the tests specified in 4.6.3.1 through 4.6.3.5 during exposure to a temperature of -40 + 5°F. Subsequently, the bulldozer shall be returned to the room ambient conditions specified in 4.3, serviced as specified in 3.3.1 and shall pass the tests specified in 4.6.3.1 through 4.6.3.5.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The bulldozer, when mounted on the applicable tank (see 1.1), is intended for use in assault on obstacles, reduction of bunkers and emergency construction operations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable drawing number, title, and date (see 3.3).
c. If first article is not required (see 3.1).
d. If marking shall be as specified by the procuring activity. Specify marking required (see 3.5).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
g. If inspection conditions shall be other than as specified (see 4.3).
h. Specify location of first article tests (see 4.4).
i. If preproduction inspection is required (see 4.4.1).
j. If initial production is not required (see 4.4.2).
k. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Subject term (key word) listing:

Kit Bulldozer
Bulldozer
Earth Moving

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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